CHMB23 Thermodynamics · Formula Sheet 热力学公式表

$\Delta U = q + w, q, w > 0$ into system · T in K · i.g. = perfect gas · rev = reversible · [ ] = condition
heat & work · entropy · third law · engines · free energy

Constants · units 常数与单位

$R = 8.314\ \mathrm{J\,K^{-1}\,mol^{-1}} = 0.08206\ \mathrm{L\,atm\,K^{-1}\,mol^{-1}}$ · $k_\mathrm{B} = R/N_\mathrm{A} = 1.381 \times 10^{-23}\ \mathrm{J\,K^{-1}}$ · $N_\mathrm{A} = 6.022 \times 10^{23}\ \mathrm{mol^{-1}}$

$p^\circ = 1\ \mathrm{bar} = 10^5\ \mathrm{Pa}$ · $1\ \mathrm{atm} = 101.325\ \mathrm{kPa} = 1.01325\ \mathrm{bar}$ · $1\ \mathrm{L\,atm} = 101.325\ \mathrm{J}$ · $1\ \mathrm{J} = 1\ \mathrm{Pa\,m^3}$

$T/\mathrm{K} = \theta/^\circ\mathrm{C} + 273.15$ · $\Delta T/\mathrm{K} = \Delta\theta/^\circ\mathrm{C}$ · SATP: $298.15\ \mathrm{K}, 1\ \mathrm{bar}, V_\mathrm{m} = 24.79\ \mathrm{L\,mol^{-1}}$

$\ln 2 = 0.693$ · $\ln\frac{3}{2} = 0.405$ · $R\ln 2 = 5.76$ · $R\ln\frac{3}{2} = 3.37\ \mathrm{J\,K^{-1}\,mol^{-1}}$

First law · work · heat 第一定律 · 功 · 热

$\Delta U = q + w$ [$q > 0$ *heat in*, $w > 0$ *work on system*] · $W_\mathrm{out} = -w$ · isolated: $q = w = 0 \Rightarrow \Delta U = 0$ · general: $E = U + E_\mathrm{kin} + E_\mathrm{pot}$

State U, H, S, A, G: $\mathrm{d}U$ exact, $\oint \mathrm{d}U = \oint \mathrm{d}H = 0$ · **path** q, w: $\delta q, \delta w$ · cycle: $q_\mathrm{net} = -w_\mathrm{net}$

$\delta w_{pV} = -p_\mathrm{ex}\mathrm{d}V$ [*rev & irrev*] · $w = -\int_{V_i}^{V_f} p_\mathrm{ex}\mathrm{d}V$ · W_out = area under $p_\mathrm{ex}(V)$ [p_gas *only if rev*]

Const p_ex $w = -p_\mathrm{ex}(V_f - V_i)$ · **free exp.** $p_\mathrm{ex} = 0 \Rightarrow w = 0$ [$q = 0$ *only if insulated*]

Rev isothermal i.g. $w = -nRT\ln\frac{V_f}{V_i} = -nRT\ln\frac{p_i}{p_f}$ · $\Delta U = 0, q = -w > 0$ [*max* W_out; *w most negative*]

Adiabatic $q = 0$: $\Delta U = w = nC_{V,\mathrm{m}}(T_f - T_i)$ · rev i.g.: $pV^\gamma, TV^{\gamma-1}$ const, $\gamma = C_p/C_V$ · **diathermic** $q \neq 0$ possible

Equilibrium movable wall $p_1 = p_2$ (same T: $V_1/V_2 = n_1/n_2$) · thermal $T_1 = T_2$ · chemical $\mu_1 = \mu_2$ · work: ordered motion, heat: random motion

Enthalpy · heat capacity 焓与热容

$H \equiv U + pV$ · $\mathrm{d}H = \mathrm{d}U + p\,\mathrm{d}V + V\mathrm{d}p$ · $\Delta H = \Delta U + \Delta(pV)$ · $H = G + TS$

$q_V = \Delta U$ [*const* V, *no other work*] · $q_p = \Delta H$ [*const* p, pV *work only*] · extra work: $\Delta H = q_p + w_\mathrm{other}$

$q_V = \Delta H$ only if $\Delta(pV) = 0$ · i.g., const V: $\Delta H - \Delta U = nR\Delta T$

$C_V = \left(\frac{\partial U}{\partial T}\right)_V$ · $C_p = \left(\frac{\partial H}{\partial T}\right)_p$ · $\Delta U = \int C_V\mathrm{d}T$ · $\Delta H = \int C_p\mathrm{d}T$

$C = nC_\mathrm{m}$: C extensive ($\mathrm{J\,K^{-1}}$), C_m intensive ($\mathrm{J\,K^{-1}\,mol^{-1}}$)

i.g.: $C_{p,\mathrm{m}} - C_{V,\mathrm{m}} = R$ · general: $C_p - C_V = TV\alpha^2/\kappa_T \geq 0$

Monatomic i.g.: $U_\mathrm{m} = \frac{3}{2}RT, C_{V,\mathrm{m}} = \frac{3}{2}R, C_{p,\mathrm{m}} = \frac{5}{2}R, \gamma = \frac{5}{3}$ · diatomic: $C_{V,\mathrm{m}} \approx \frac{5}{2}R, \gamma = \frac{7}{5}$

i.g.: $U = U(T)$ · $\left(\frac{\partial U}{\partial V}\right)_T = T\left(\frac{\partial p}{\partial T}\right)_V - p = 0$

Bomb calorimeter: $q_V = \Delta U \approx -C_\mathrm{cal}\Delta T$ · const-p calorimeter: ΔH

Thermochemistry 热化学

Exothermic $\Delta_\mathrm{r}H < 0$ ($q_p < 0$) · endothermic $\Delta_\mathrm{r}H > 0$ · reverse reaction: $-\Delta_\mathrm{r}H$

$\Delta_\mathrm{r}H = \Delta_\mathrm{r}U + \Delta\nu_\mathrm{gas}RT$ [*i.g.; condensed* pV *neglected*] · $\Delta H \approx \Delta U$ if $\Delta n_\mathrm{gas} = 0$

$\Delta_\mathrm{r}X^\circ = \sum_\mathrm{prod}\nu X^\circ - \sum_\mathrm{react}\nu X^\circ$ [$X = \Delta_\mathrm{f}H,\ \Delta_\mathrm{f}G,\ S_\mathrm{m},\ C_{p,\mathrm{m}}$] · elements [*ref. state*]: $\Delta_\mathrm{f}H^\circ = \Delta_\mathrm{f}G^\circ = 0$

Kirchhoff $\Delta_\mathrm{r}H^\circ(T_2) = \Delta_\mathrm{r}H^\circ(T_1) + \int_{T_1}^{T_2}\Delta_\mathrm{r}C_p^\circ\mathrm{d}T \approx \Delta_\mathrm{r}H^\circ(T_1) + \Delta_\mathrm{r}C_p^\circ\,(T_2 - T_1)$ [$\approx$: *const* $\Delta_\mathrm{r}C_p^\circ$; *phase change: add* $\Delta_\mathrm{trs}H$]

Standard state $^\circ$: $p^\circ = 1\ \mathrm{bar}$, any stated T [*not fixed at 298.15 K*] · $\Delta_\mathrm{fus}H(\mathrm{H_2O}) = +6.01\ \mathrm{kJ\,mol^{-1}}$

Gases · molecular picture 气体与分子图像

$pV = nRT = Nk_\mathrm{B}T$ · $p_J = x_J p$, $\sum_J p_J = p$ · i.g. mixture: $p_J = n_J RT/V$

$Z = \frac{pV_\mathrm{m}}{RT} = 1 + \frac{B}{V_\mathrm{m}} + \frac{C}{V_\mathrm{m}^2} + \cdots$ · $p \to 0$: $V_\mathrm{m} \to \infty$, $Z \to 1$

Kinetic model [*point mass, random, elastic, no forces*]: $pV = \frac{1}{3}Nm\langle v^2\rangle$ · $\frac{1}{2}m\langle v^2\rangle = \frac{3}{2}k_\mathrm{B}T$

$v_\mathrm{rms} = \sqrt{3RT/M}$ · $\bar{v} = \sqrt{8RT/\pi M}$ · $v_\mathrm{mp} = \sqrt{2RT/M}$ · $v_\mathrm{mp} < \bar{v} < v_\mathrm{rms}$ [M *in* $\mathrm{kg\,mol^{-1}}$] · $\mathrm{N_2}$, 300 K: $517\ \mathrm{m\,s^{-1}}$

Maxwell–Boltzmann $f(v) \propto v^2 e^{-Mv^2/2RT}$ · $T\uparrow$: peak to higher v, broader, lower, area $= 1$

Equipartition: $\frac{1}{2}k_\mathrm{B}T$ per quadratic term [*classical*, $k_\mathrm{B}T \gg \Delta\epsilon$] · translation $U_\mathrm{m} = \frac{3}{2}RT$

Vibration $\langle\epsilon\rangle = \frac{\hbar\omega}{2} + \frac{\hbar\omega}{e^{\hbar\omega/k_\mathrm{B}T}-1}$ [$k_\mathrm{B}T \lesssim \hbar\omega$: *frozen out*]

Boltzmann $\frac{N_j}{N_i} = \frac{g_j}{g_i}e^{-(\epsilon_j-\epsilon_i)/k_\mathrm{B}T}$ · $T\uparrow$: $N_j/N_i\uparrow$

Box $E = \frac{h^2}{8mL^2}(n_x^2 + n_y^2 + n_z^2) \propto V^{-2/3}$ · $q_\mathrm{trans} = V/\Lambda^3 \propto V$ [$V\uparrow$: *denser levels*, $W\uparrow$]

Entropy · second law 熵与第二定律

$\mathrm{d}S = \frac{\delta q_\mathrm{rev}}{T}$ [*irrev: rev path, same end states*] · $S = k_\mathrm{B}\ln W$ · $S = -k_\mathrm{B}\sum_i p_i\ln p_i$ · $W = 1 \Rightarrow S = 0$

Clausius $\mathrm{d}S \geq \frac{\delta q}{T}$ [$=$ *only rev*] · isolated: $\Delta S = S_\mathrm{gen} \geq 0$ [> 0 *irrev*, $= 0$ *rev*]

Balance $\Delta S = \int\frac{\delta q}{T_\mathrm{b}} + S_\mathrm{gen}$ [*no work term*] · adiabatic stirring: $q = 0, \Delta S = S_\mathrm{gen} > 0$

$\Delta S_\mathrm{tot} = \Delta S_\mathrm{sys} + \Delta S_\mathrm{surr}$: > 0 spontaneous · $= 0$ equilibrium, rev · < 0 impossible

Reservoir $\Delta S_\mathrm{surr} = \frac{q_\mathrm{surr}}{T_\mathrm{surr}} = -\frac{q_\mathrm{sys}}{T_\mathrm{surr}}$ [*actual q; rev or irrev*] · const p: $\Delta S_\mathrm{surr} = -\frac{\Delta H_\mathrm{sys}}{T_\mathrm{surr}}$ [pV *work only*]

$\Delta S = q_\mathrm{rev}/T$ [*const* T]: same $q, T\downarrow \Rightarrow\ \Delta S\uparrow$ · 100 J: $0.400\ \mathrm{J\,K^{-1}}$ (250 K), $0.200\ \mathrm{J\,K^{-1}}$ (500 K)

Rev: $S_\mathrm{gen} = 0$ [*ideal limit, infinite time*] · quasistatic + friction: $S_\mathrm{gen} > 0$

Calculating ΔS 熵变计算

i.g.: $\Delta S = nC_{V,\mathrm{m}}\ln\frac{T_f}{T_i} + nR\ln\frac{V_f}{V_i} = nC_{p,\mathrm{m}}\ln\frac{T_f}{T_i} - nR\ln\frac{p_f}{p_i}$

Isothermal i.g. $\Delta S = nR\ln\frac{V_f}{V_i}$ [*rev or irrev*] · rev: $\Delta S_\mathrm{surr} = -nR\ln\frac{V_f}{V_i}, \Delta S_\mathrm{tot} = 0$ · free exp.: $\Delta S_\mathrm{surr} = 0, \Delta S_\mathrm{tot} > 0$

Compression $\Delta S < 0$ · 1 mol, V halved: $-R\ln 2 = -5.76\ \mathrm{J\,K^{-1}}$

Heating $\Delta S = \int_{T_i}^{T_f}\frac{C_p}{T}\mathrm{d}T = C_p\ln\frac{T_f}{T_i}$ [*const p; const* C_p] · C_V [*const* V] · $C_p\Delta T = q_p$, not ΔS

Rev adiabatic: $\Delta S = 0$ (isentropic) · irrev adiabatic: $\Delta S > 0$

At T_trs $\Delta_\mathrm{trs}S = \frac{\Delta_\mathrm{trs}H}{T_\mathrm{trs}}, \Delta S_\mathrm{surr} = -\frac{\Delta_\mathrm{trs}H}{T_\mathrm{trs}}, \Delta S_\mathrm{tot} = 0$ · freezing 1 mol water: $\frac{-6010\ \mathrm{J}}{273.15\ \mathrm{K}} = -22.0\ \mathrm{J\,K^{-1}}$

Trouton $\Delta_\mathrm{vap}S \approx 85\ \mathrm{J\,K^{-1}\,mol^{-1}}$ (80–90) $\gg \Delta_\mathrm{fus}S \approx$ 8–25 · $S_\mathrm{m}(\mathrm{g}) > S_\mathrm{m}(\mathrm{l}) > S_\mathrm{m}(\mathrm{s})$

Ideal mixing $\Delta_\mathrm{mix}S = -nR\sum_i x_i\ln x_i > 0$ · $\Delta_\mathrm{mix}G = nRT\sum_i x_i\ln x_i$ · $\Delta_\mathrm{mix}H = \Delta_\mathrm{mix}V = 0$ · $\mu_i = \mu_i^* + RT\ln x_i$

Power P: $Q = Pt, \Delta S_\mathrm{surr} = Q/T$ (100 W, 1 day, 293 K: $+29.5\ \mathrm{kJ\,K^{-1}}$)

$\Delta_\mathrm{r}S^\circ = \sum\nu S_\mathrm{m}^\circ(\mathrm{prod}) - \sum\nu S_\mathrm{m}^\circ(\mathrm{react})$ [*absolute* S_m°; *elements* $S_\mathrm{m}^\circ \neq 0$]

Third law · low T 第三定律 · 低温

$S \to 0$ as $T \to 0$ [*perfect crystal, unique ground state*] · $C \to 0$ as $T \to 0$ (yet $S(0) > 0$ possible)

$S_\mathrm{m}(T) = \int_0^T\frac{C_{p,\mathrm{m}}}{T'}\mathrm{d}T' + \sum\frac{\Delta_\mathrm{trs}H}{T_\mathrm{trs}} + S_\mathrm{res}$ [*area under* C_p/T *vs* T]

Debye [$T \ll \Theta_\mathrm{D}$, *nonmetal*] $C_{V,\mathrm{m}} = \frac{12\pi^4}{5}R\left(\frac{T}{\Theta_\mathrm{D}}\right)^3 = aT^3$ · $S_\mathrm{m} = \frac{aT^3}{3} = \frac{1}{3}C_{p,\mathrm{m}}$ · $C = aT^r \Rightarrow S = C/r$ · metals $+\gamma_\mathrm{el}T$

Residual $S_\mathrm{res} = k_\mathrm{B}\ln W_0$ · CO: $W_0 = 2^N, R\ln 2 = 5.76$ (expt 4–5) · ice: $W_0 = (3/2)^N, R\ln\frac{3}{2} = 3.4\ \mathrm{J\,K^{-1}\,mol^{-1}}$ [*proton disorder*] · ice XI ordered

Engines · fridges · heat pumps 热机 · 制冷 · 热泵

Carnot $\frac{Q_h}{T_h} = \frac{Q_c}{T_c}$ · $W = Q_h - Q_c$ · $\eta = \frac{W}{Q_h} = 1 - \frac{Q_c}{Q_h} \leq 1 - \frac{T_c}{T_h}$ [*K only*] · $T_c > 0 \Rightarrow \eta < 1$

$\left(\frac{\partial\eta}{\partial T_h}\right)_{T_c} = \frac{T_c}{T_h^2} > 0$ · $\left(\frac{\partial\eta}{\partial T_c}\right)_{T_h} = -\frac{1}{T_h} < 0$ · $\eta = f(T_c/T_h)$, not ΔT: $(600, 300) \to 0.50$, $(900, 600) \to 0.333$ · $(473, 293) \to 0.38$

Fridge $\mathrm{COP_{cool}} = \frac{Q_c}{W} \leq \frac{T_c}{T_h - T_c}$ · **heat pump** $\mathrm{COP_{warm}} = \frac{Q_h}{W} \leq \frac{T_h}{T_h - T_c} = \mathrm{COP_{cool}} + 1$ · COP > 1 possible

Kelvin-Planck 1 reservoir: $\Delta S_\mathrm{tot} = -\frac{Q_h}{T_h} < 0$ ✗ · 2 reservoirs: $-\frac{Q_h}{T_h} + \frac{Q_c}{T_c} \geq 0$ · fridge: $\frac{Q_h}{T_h} - \frac{Q_c}{T_c} \geq 0, W = Q_h - Q_c > 0$ · cycle: $\Delta S_\mathrm{engine} = 0$

Free energy · spontaneity 自由能与自发判据

$A = U - TS$ · $G = H - TS = A + pV$ · $\Delta G - \Delta A = \Delta(pV)$, i.g. $= nR\Delta T \Rightarrow$ isothermal i.g.: $\Delta G = \Delta A$

$\mathrm{d}U = T\mathrm{d}S - p\mathrm{d}V$ · $\mathrm{d}H = T\mathrm{d}S + V\mathrm{d}p$ · $\mathrm{d}A = -S\mathrm{d}T - p\mathrm{d}V$ · $\mathrm{d}G = -S\mathrm{d}T + V\mathrm{d}p$

Maxwell $\left(\frac{\partial T}{\partial V}\right)_S = -\left(\frac{\partial p}{\partial S}\right)_V$ · $\left(\frac{\partial T}{\partial p}\right)_S = \left(\frac{\partial V}{\partial S}\right)_p$ · $\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial p}{\partial T}\right)_V$ · $\left(\frac{\partial S}{\partial p}\right)_T = -\left(\frac{\partial V}{\partial T}\right)_p$

Spontaneous: $\Delta A < 0$ [*const* T, V] · $\Delta G < 0$ [*const* T, p] · $= 0$ equilibrium [pV *work only*] · nonspontaneous: work input

$\Delta G = \Delta H - T\Delta S = -T\Delta S_\mathrm{tot}$ [*const* T, p] · $\Delta A = -T\Delta S_\mathrm{tot}$ [*const* T, V]

Max work $W_\mathrm{out}^\mathrm{max} = -\Delta A$ (total) [*const* T] · $W_\mathrm{non\text{-}}^\mathrm{max}{}_{pV} = -\Delta G$ (e.g. cell) [*const* T, p] · on system: $w_\mathrm{rev} = \Delta A$, $w_{\text{non-}pV,\mathrm{rev}} = \Delta G$

$\Delta H < 0, \Delta S > 0$: all T · $\Delta H > 0, \Delta S < 0$: no T · $\Delta H < 0, \Delta S < 0$: low T · $\Delta H > 0, \Delta S > 0$: high T · $T^* = \Delta H/\Delta S$

$\Delta_\mathrm{r}G = \Delta_\mathrm{r}G^\circ + RT\ln Q$ [*direction*] · $\Delta_\mathrm{r}G^\circ = -RT\ln K = \Delta_\mathrm{r}H^\circ - T\Delta_\mathrm{r}S^\circ$

$\left(\frac{\partial G}{\partial T}\right)_p = -S$ · $\left(\frac{\partial G}{\partial p}\right)_T = V$ · isothermal i.g.: $\Delta G = nRT\ln\frac{p_f}{p_i}$ · condensed: $\Delta G_\mathrm{m} \approx V_\mathrm{m}\Delta p$ · $\left(\frac{\partial(G/T)}{\partial T}\right)_p = -\frac{H}{T^2}$

考点速记 · Key traps

✓ $\Delta S_\mathrm{sys} < 0$ [$\Delta S_\mathrm{surr} > |\Delta S_\mathrm{sys}|$] · ✓ endothermic + spontaneous ($T\Delta S > \Delta H$, ice melting) · spontaneous ≠ fast

$q_V = \Delta H, \Delta H = \Delta U, \Delta G = \Delta A$: only if $\Delta(pV) = 0$ · i.g. isothermal ✓ (even with $\Delta p, \Delta V \neq 0$)

Max output $-\Delta A$, $-\Delta G$ (signs) · $-\Delta A$ total work · $-\Delta G$ non-pV work only

$S(0) > 0$ possible (CO, ice, glass) though $C \to 0$ · $S_\mathrm{m}^\circ(\mathrm{element}) \neq 0$ · $\Delta_\mathrm{f}H^\circ(\mathrm{element}) = 0$

Free exp. ($p_\mathrm{ex} = 0$): $w = 0$ · insulated i.g.: $q = 0, \Delta U = \Delta T = 0, \Delta S = nR\ln\frac{V_f}{V_i} > 0$

Kirchhoff linear form: const $\Delta_\mathrm{r}C_p^\circ$ only · Debye $S = C/3$, ✗ $S = C$ · $\Delta S = \int C\,\mathrm{d}T/T$, ✗ $C\Delta T$

η, COP, $q/T, pV = nRT$: K only, ✗ °C · $p^\circ = 1\ \mathrm{bar} \neq 1\ \mathrm{atm}$ · standard state: any T

Adiabatic ≠ isothermal · quasistatic ≠ reversible · ideal mixture: equal interactions, not necessarily zero

Both ✓: $-nRT\ln\frac{V_2}{V_1} = -nRT\ln\frac{p_1}{p_2}$ · $H = U + pV = G + TS$ · $\delta w = -p_\mathrm{ex}\mathrm{d}V$ for rev & irrev